VIDEO FUNDAMENTALS



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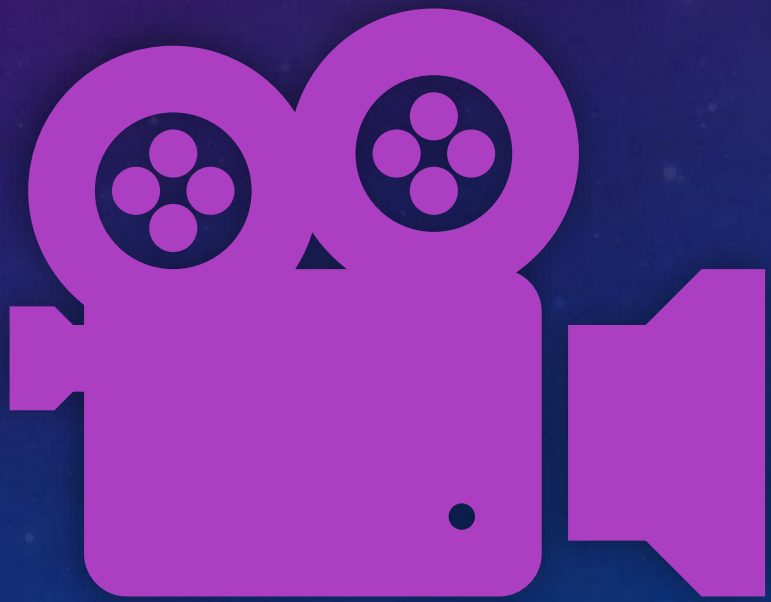
WHAT IS VIDEO?

- Generally, a video is used to generate a steady source of still pictures as it is a series of electronic signals, which simulate movement.
- Videos can be used for education, entertainment, or other purposes, which use pictures, graphics, or text.
- In modern times, there are various websites or webpages that contain streaming or downloadable video, which can be watched by the users on their computer or other similar devices.

FORMATS OF VIDEO FILES?

There are numerous video file formats available that can be used for videos.

Eg: **.3GP:**
.MP4
.AVI
.WebM
.MOV
.WMV
.FLV



TYPES OF VIDEO

There are many types video.

Eg:

- **vlog**
- **behind the scenes**
- **interview**
- **product review**
- **webinar**

CALCULATING THE VIDEO FILE SIZE

In order to complete this calculation, you will need to find the following about the image.

- The Duration of the video(Time)
- How many frames per second (fps)
- The amount of Horizontal Pixels
- The amount of Vertical Pixels
- Bit Depth of the image – the number of bits per pixel. More pixels = More colors

CALCULATING THE VIDEO FILE SIZE

Number of Frames

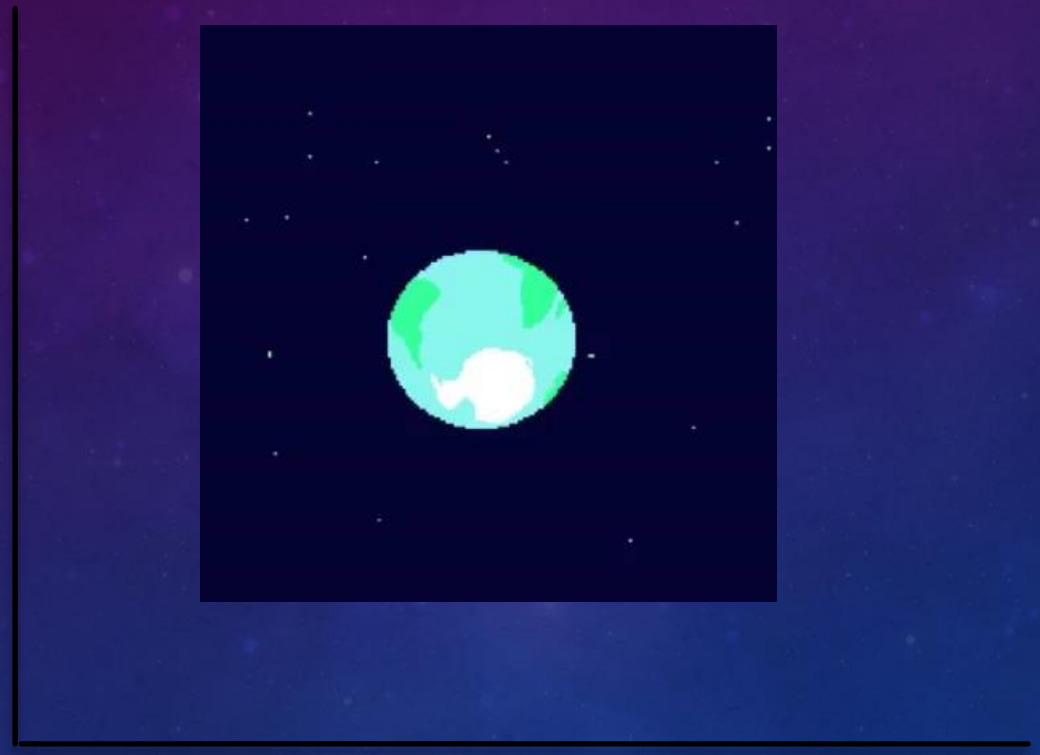
Frame Rate * Time in Seconds

File Size of a Frame

$$\frac{\text{Horizontal} * \text{Vertical} * \text{Bit Depth}}{8 * 1024}$$

CALCULATING THE VIDEO FILE SIZE

600 pixels



800 pixels

Bit Depth = 6

Frames per second (fps) = 30

Time = 3 Seconds

CALCULATING THE VIDEO FILE SIZE

Number of Frames

Frame Rate * Time in Seconds

30 Frames per second * 3 Seconds = 90

File Size of a Frame

$$\frac{\text{Horizontal} * \text{Vertical} * \text{Bit Depth}}{8 * 1024} = \frac{800 \text{ pixels} * 600 \text{ pixels} * 6}{8 * 1024} = 351.56 \text{ KB}$$

CALCULATING THE VIDEO FILE SIZE

=90 * 351.56 Kb

=31640.63 KB / 1024

=30.9 MB

THANK YOU